

BEGUM ROKEYA UNIVERSITY, RANGPUR

ASSIGNMENT ON

COURSE CODE: STAT-2205

COURSE TITLE: PROGRAMMING WITH R & PYTHON

SUBMITTED BY	SUBMITTED TO
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1 Introduction

This report presents the analysis of coach ticket prices using both R and Python programming languages.

FILE : FLIGHT_DATA

2 Question 01

What do coach ticket prices look like? What are the high and low values? What would be considered the average? Does \$500 seem like a good price for a coach ticket?

2.1 R Code

```
getwd()
list.files()

setwd("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")

.libPaths()
.libPaths("C:/Users/Eva/Downloads/BRUR_2026(STAT-2205)/R_PROGRAMMING")

d <- read.csv(file.choose(), header = TRUE)

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

summary(sample_df$coach_price)

sd(sample_df$coach_price)
```

2.2 R Output

```
summary(sample_df$coach_price)
Min.    1st Qu.    Median    Mean    3rd Qu.    Max.
44.41    331.45    379.98    376.14    425.94    588.34
sd(sample_df$coach_price)
67.69549
```

2.3 R Interpretation

Coach ticket prices vary widely, ranging from \$44.41 to \$588.34. The average (mean) coach ticket price is approximately \$376.14, while the median price is \$379.98, indicating that a typical coach ticket costs around \$380.

Most coach ticket prices fall between \$331.45 and \$425.94, meaning that the majority of prices are in the mid-\$300 to low-\$400 range. The standard deviation of \$67.70 indicates moderate variation in prices.

A coach ticket priced at \$500 would generally be considered expensive because it is much higher than the average ticket price and closer to the maximum observed value.

2.4 Python Code

```
import os
print(os.getcwd())

os.chdir("C:/Users/Eva/Downloads/BRUR_2026(STAT2205)/R_PROGRAMMING")
os.listdir()

import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print(sample_df['coach_price'].describe())
```

2.5 Python Output

count	10063.000000
mean	377.725740
std	67.229885
min	130.500000
25%	332.377500
50%	381.540000
75%	427.965000
max	562.925000

2.6 PYTHON Interpretation

The coach ticket prices range from \$130.50 to \$562.93. The average coach ticket price is approximately \$377.73, while the median price is \$381.54, indicating that a typical coach ticket usually costs around \$380.

Most coach ticket prices fall between \$332.38 and \$427.97. The standard deviation of \$67.23 shows moderate variation in ticket prices.

A \$500 coach ticket would generally be considered expensive because it is much higher than the average price and close to the maximum observed value.

3 Question 02

What are the high, low, and average prices for 8-hour-long flights? Does a \$500 ticket seem more reasonable than before?

3.1 R Code

```
install.packages("tidyverse")

library(tidyverse)

eight_hr <- sample_df %>% filter(hours == 8)

cat("Number of 8-hour flights:", nrow(eight_hr), "\n")

summary(eight_hr$coach_price)

sd(eight_hr$coach_price)
```

3.2 R Output

Number of 8-hour flights: 211

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
215.20	388.00	426.20	429.10	479.00	588.30

Standard Deviation:
66.85

3.3 R Interpretation

For 8-hour-long flights, coach ticket prices range from approximately \$215 to \$588. The average coach ticket price is about \$429, indicating that longer flights are generally more expensive than average flights overall.

The standard deviation is approximately \$66, which shows moderate variation in ticket prices. Most ticket prices are distributed around the average value, meaning that fares are relatively consistent for long-duration flights.

A \$500 coach ticket is above the average price but still falls within the normal range of prices for 8-hour flights. Therefore, compared to the previous analysis, a \$500 ticket now appears more reasonable for a long-duration flight.

3.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)
```

```

sample_df = df.sample(n=10063, random_state=10063).copy()

eight_hr = sample_df[sample_df['hours'] == 8]

print("Number of 8-hour flights:", len(eight_hr))

print(eight_hr['coach_price'].describe())

```

3.5 Python Output

Number of 8-hour flights: 211

```

count      211.000000
mean       426.952299
std         57.840730
min        235.745000
25%        393.770000
50%        426.505000
75%        467.630000
max        562.925000

```

3.6 PYTHON Interpretation

Coach ticket prices for 8-hour flights are generally centered around the mid-\$400 range. Prices range from approximately \$235.75 to \$562.93. The average price is about \$426.95, while the median price is \$426.51, indicating a fairly balanced distribution of ticket prices.

Most ticket prices fall between \$393.77 and \$467.63, showing that the majority of fares are concentrated within a relatively narrow range. The standard deviation of approximately \$57.84 indicates moderate variation in prices.

A \$500 ticket is higher than the average price but still falls within the upper range of normal ticket prices for 8-hour flights. Therefore, it appears more reasonable than it did in the overall coach ticket price analysis.

4 Question 03

How are flight delay times distributed? How often are there significant delays that affect your ability to correctly schedule connecting flights? What kinds of delays are typical?

4.1 R Code

```

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

```

```

cat("=== Flight Delay Distribution ===\n")

print(summary(sample_df$delay))

cat("Standard Deviation:", round(sd(sample_df$delay), 2), "\n\n")

cat("Delays > 30 minutes:", sum(sample_df$delay > 30),
    paste0("(", round(mean(sample_df$delay > 30)*100, 2), "%)"), "\n")

cat("Delays > 60 minutes:", sum(sample_df$delay > 60), "\n")

cat("\nMost Common Delays:\n")

print(head(sort(table(sample_df$delay), decreasing = TRUE), 10))

```

4.2 R Output

```

Min. 1st Qu. Median Mean 3rd Qu. Max.
0.00  9.00 10.00 12.74 13.00 1507.00

```

```
Standard Deviation: 33.9
```

```
Delays > 30 minutes: 490 (4.87%)
```

```
Delays > 60 minutes: 5
```

```
Most Common Delays:
```

```

10  9  11  12  8  0  13  7  14  6
1524 1339 1275 931 895 808 506 467 261 189

```

4.3 R Interpretation

Flight delay times are mostly concentrated around small delays, with most flights delayed between 9 and 13 minutes. The median delay is 10 minutes and the average delay is approximately 12.74 minutes. However, the maximum delay is 1507 minutes, indicating that a few flights experienced extremely large delays. This suggests that the distribution is positively skewed (right-skewed).

Significant delays that could affect connecting flights are relatively uncommon. About 490 flights, or 4.87% of all flights, experienced delays greater than 30 minutes, while delays longer than 60 minutes occurred in only 5 flights.

Typical delays are short, especially between 8 and 12 minutes, with a 10-minute delay being the most common. Therefore, most passengers experience only minor delays, while serious disruptions happen infrequently.

4.4 Python Code

```
import pandas as pd
```

```

import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== Flight Delay Distribution ===")

print(sample_df['delay'].describe())

print("\nSignificant Delays:")

print("Delays > 30 minutes:", (sample_df['delay'] > 30).sum(),
      f"({round((sample_df['delay'] > 30).mean()*100, 2)}%)")

print("Delays > 60 minutes:", (sample_df['delay'] > 60).sum())

print("\nMost Common Delays:")

print(sample_df['delay'].value_counts().head(10))

```

4.5 Python Output

=== Flight Delay Distribution ===

```

count    10063.000000
mean       13.056643
std        40.118008
min         0.000000
25%         9.000000
50%        10.000000
75%        13.000000
max       1560.000000

```

Significant Delays:

Delays > 30 minutes: 502 (4.99%)

Delays > 60 minutes: 7

Most Common Delays:

```

Delay
10    1443
9     1303
11    1290

```

12	968
8	915
0	811
13	538
7	502
14	238
6	190

4.6 PYTHON Interpretation

Flight delay times are mostly small, with the distribution concentrated around 9–13 minutes. The average delay is about 13 minutes, while the median delay is 10 minutes, showing that typical delays are relatively short. However, the maximum delay is 1560 minutes, indicating a few extremely large delays that make the distribution strongly right-skewed.

Significant delays that may affect connecting flights are not very common. About 502 flights (4.99%) experienced delays greater than 30 minutes, while only 7 flights had delays over 60 minutes. This suggests that serious disruptions are relatively rare.

The most typical delays are between 8 and 12 minutes, with a 10-minute delay being the most frequent. Therefore, most passengers are likely to experience only minor delays, while major delays occur infrequently.

5 Question 04

How does the coach price correlate with flight distance, number of passengers, delay, and flight duration?

5.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_cols <- c("miles", "passengers", "delay",
                  "hours", "coach_price")

cor_matrix <- cor(sample_df[numeric_cols])

print(round(cor_matrix[, "coach_price"], 4))
```

5.2 R Output

miles	passengers	delay	hours	coach_price
0.3381	0.1720	0.0166	0.3317	1.0000

5.3 R Interpretation

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3381, while the correlation with hours is 0.3317. This suggests that flights covering longer distances and taking more time generally have higher coach ticket prices.

The correlation between coach price and number of passengers is weak (0.1720), meaning passenger count has only a slight relationship with ticket price.

The correlation between coach price and delay is extremely weak (0.0166), indicating that delays have almost no effect on coach ticket prices.

Overall, coach ticket prices are mostly related to flight distance and duration, while passenger numbers and delays show very little influence.

5.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063, random_state=10063).copy()

numeric_cols = ['miles', 'passengers', 'delay',
                'hours', 'coach_price']

correlation = sample_df[numeric_cols].corr()['coach_price'].round(4)

print(correlation)
```

5.5 Python Output

miles	0.3408
passengers	0.1667
delay	0.0129
hours	0.3344
coach_price	1.0000

5.6 PYTHON Interpretation

Coach ticket price has a moderate positive correlation with both flight distance and flight duration. The correlation between coach price and miles is 0.3408, while the correlation with hours is 0.3344. This means that longer flights and flights covering greater distances generally tend to have higher coach prices.

The correlation between coach price and number of passengers is relatively weak (0.1667), suggesting that passenger count has only a small effect on ticket prices.

The correlation between coach price and delay is extremely weak (0.0129), indicating that flight delays have almost no relationship with coach ticket prices.

Overall, coach prices are mainly influenced by how far and how long the flight is, while delays and passenger numbers have little impact.

6 Question 05

What is the relationship between coach and first-class prices? Do flights with higher coach prices always have higher first-class prices as well?

6.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

correlation <- cor(sample_df$coach_price,
                    sample_df$firstclass_price)

cat("Correlation between Coach and First Class Price:",
    round(correlation, 4), "\n")

cat("\nCoach Price Mean      :",
    round(mean(sample_df$coach_price), 2), "\n")

cat("First Class Price Mean:",
    round(mean(sample_df$firstclass_price), 2), "\n")
```

6.2 R Output

```
Correlation between Coach and First Class Price: 0.7556
```

```
Coach Price Mean      : 376.14
```

```
First Class Price Mean: 1453.01
```

6.3 R Interpretation

The correlation between coach price and first-class price is 0.7556, which indicates a strong positive relationship between the two variables. This means that flights with higher coach prices generally tend to have higher first-class prices as well, while flights with lower coach prices tend to have lower first-class prices.

However, the relationship is not perfect. Therefore, higher coach prices do not always guarantee higher first-class prices. Some exceptions may occur because of airline pricing strategies, travel demand, and route differences.

The average coach ticket price is \$376.14, while the average first-class ticket price is \$1453.01, showing that first-class tickets are significantly more expensive on average.

6.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

correlation = sample_df['coach_price'].corr(
    sample_df['firstclass_price'])

print(f"Correlation between Coach and First Class Price:
      {correlation:.4f}")

print("\nSummary Statistics:")

print("Coach Price Mean      :",
      round(sample_df['coach_price'].mean(), 2))

print("First Class Price Mean:",
      round(sample_df['firstclass_price'].mean(), 2))
```

6.5 Python Output

Correlation between Coach and First Class Price: 0.7547

Summary Statistics:

Coach Price Mean : 377.73

First Class Price Mean: 1457.39

6.6 PYTHON Interpretation

The correlation between coach price and first-class price is 0.7547, which indicates a strong positive relationship between the two variables. This means that, in general, when coach prices increase, first-class prices also tend to increase, and when coach prices decrease, first-class prices also tend to decrease.

However, this relationship is not perfect. Therefore, flights with higher coach prices do not always have higher first-class prices, although they usually follow the same pricing trend.

The mean coach ticket price is \$377.73, while the mean first-class ticket price is \$1457.39, indicating that first-class tickets are significantly more expensive on average.

7 Question 06

What is the relationship between coach prices and in-flight features: in-flight meal, in-flight entertainment, and in-flight WiFi? Which features are associated with the highest increase in price?

7.1 R Code

```
features <- c("inflight_meal",
             "inflight_entertainment",
             "inflight_wifi")

for (f in features) {

  cat("\n=== ", gsub("_", " ", f), " ===\n")

  stats <- sample_df %>%
    group_by(.data[[f]]) %>%
    summarise(
      count = n(),
      mean_price = mean(coach_price),
      median_price = median(coach_price),
      .groups = "drop"
    )

  yes_mean <- stats %>%
    filter(.data[[f]] == "Yes") %>%
    pull(mean_price)

  no_mean <- stats %>%
    filter(.data[[f]] == "No") %>%
    pull(mean_price)

  premium <- yes_mean - no_mean

  stats_print <- stats %>%
    mutate(across(where(is.numeric),
      ~sprintf("%.2f", .x)))

  print(stats_print)

  cat("Price Premium (Yes vs No): $",
    sprintf("%.2f", premium), "\n")
}
```

7.2 R Output

```
=== inflight meal ===
```

No	7091.00	369.92	373.15
Yes	2972.00	390.96	395.40

Price Premium (Yes vs No): \$21.03

=== inflight entertainment ===

No	2008.00	320.32	325.50
Yes	8055.00	390.05	394.06

Price Premium (Yes vs No): \$69.72

=== inflight wifi ===

No	1024.00	314.89	322.42
Yes	9039.00	383.07	386.83

Price Premium (Yes vs No): \$68.18

7.3 R Interpretation

Coach prices are generally higher when additional in-flight services are available.

Flights offering in-flight meals have an average coach price of approximately \$390.96, compared to \$369.92 for flights without meals. The price premium is about \$21.03, indicating only a small increase in price.

Flights with in-flight entertainment have a much higher average coach price (\$390.05) compared to flights without entertainment (\$320.32). The price premium is approximately \$69.72, showing a strong positive relationship between entertainment services and higher ticket prices.

Similarly, flights with in-flight WiFi have an average coach price of \$383.07, while flights without WiFi average \$314.89. The price premium is approximately \$68.18, indicating another strong relationship with higher prices.

Among all the features, in-flight entertainment has the highest association with increased coach ticket prices, followed closely by in-flight WiFi. In contrast, in-flight meals have only a relatively minor effect on ticket prices.

7.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()
```

```

features = ['inflight_meal',
            'inflight_entertainment',
            'inflight_wifi']

for feature in features:

    print(f"\n=== {feature.replace('_', ' ').title()} ===")

    stats = sample_df.groupby(feature)['coach_price'].agg(
        ['count', 'mean', 'median']).round(2)

    print(stats)

    premium = stats.loc['Yes', 'mean'] - stats.loc['No', 'mean']

    print(f"Price Premium (Yes vs No): ${premium:.2f}")

```

7.5 Python Output

=== Inflight Meal ===

	count	mean	median
No	7083	372.15	375.23
Yes	2980	390.98	397.56

Price Premium (Yes vs No): \$18.83

=== Inflight Entertainment ===

	count	mean	median
No	1954	320.49	325.03
Yes	8109	391.52	396.66

Price Premium (Yes vs No): \$71.03

=== Inflight Wifi ===

	count	mean	median
No	966	315.14	317.84
Yes	9097	384.37	388.53

Price Premium (Yes vs No): \$69.23

7.6 PYTHON Interpretation

The relationship between coach prices and in-flight services shows that flights offering additional features generally have higher ticket prices.

Flights with in-flight meals have a slightly higher average coach price (\$390.98) compared to flights without meals (\$372.15). The price premium is only \$18.83, indicating a relatively weak relationship.

Flights offering in-flight entertainment have a substantially higher average coach price (\$391.52) compared to flights without entertainment (\$320.49). The price premium is approximately \$71.03, showing a strong positive association.

Flights with in-flight WiFi also have higher average coach prices (\$384.37) compared to those without WiFi (\$315.14). The price premium is approximately \$69.23, which is also a strong positive association.

Among the three features, in-flight entertainment has the largest impact on increasing coach ticket prices, followed closely by in-flight WiFi. In-flight meals have the smallest effect on price.

Overall, entertainment and WiFi are the features most strongly associated with higher coach ticket prices.

8 Question 07

How does the number of passengers change in relation to the length of flights?

8.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("Overall Average Passengers:",
    round(mean(sample_df$passengers), 2), "\n\n")

cat("Average Passengers by Flight Duration:\n")

sample_df %>%
  group_by(hours) %>%
  summarise(avg_passengers = mean(passengers)) %>%
  mutate(avg_passengers = sprintf("%.2f",
    avg_passengers)) %>%
  print()
```

8.2 R Output

Overall Average Passengers: 207.74

Average Passengers by Flight Duration:

hours	avg_passengers
-------	----------------

1	207.74
---	--------

2	208.26
3	207.39
4	207.84
5	207.73
6	206.90
7	207.73
8	207.81

8.3 R Interpretation

The average number of passengers is approximately 207.74, and the values remain very stable across all flight durations from 1 to 8 hours. The average passenger count ranges only from 206.90 to 208.26.

There is no clear increasing or decreasing trend as flight duration increases. This suggests that flight length does not significantly affect the number of passengers.

Overall, passenger demand appears to remain stable regardless of flight duration, indicating that other factors such as route popularity, scheduling, or ticket pricing may have a stronger influence on passenger numbers.

8.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

print("Overall Average Passengers:",
      round(sample_df['passengers'].mean(), 2))

print("\nAverage Passengers by Flight Duration (hours):")

print(sample_df.groupby('hours')['passengers']
      .mean().round(2))
```

8.5 Python Output

Overall Average Passengers: 207.81

Average Passengers by Flight Duration (hours):

hours	
1	208.06
2	208.20

3	207.32
4	207.89
5	208.03
6	206.88
7	208.31
8	205.73

8.6 PYTHON Interpretation

The average number of passengers is approximately 207.81. Across different flight durations ranging from 1 to 8 hours, the average number of passengers remains highly consistent, varying only between 205.73 and 208.31.

There is no clear upward or downward trend as flight duration increases. This indicates that the length of a flight does not significantly influence the number of passengers.

Overall, passenger demand appears to remain stable regardless of flight duration, suggesting that factors such as pricing, route popularity, and scheduling may be more important in determining passenger numbers.

9 Question 08

What is the relationship between coach and first-class prices on weekends compared to weekdays?

9.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

cat("=== Coach Price: Weekend vs Weekday ===\n")

cat("Weekend Coach Mean      :",
    round(mean(sample_df[sample_df$weekend == "Yes", ]$coach_price), 2),
    "\n")

cat("Weekday Coach Mean      :",
    round(mean(sample_df[sample_df$weekend == "No", ]$coach_price), 2),
    "\n\n")

cat("=== First Class Price: Weekend vs Weekday ===\n")

cat("Weekend First Class Mean:",
    round(mean(sample_df[sample_df$weekend == "Yes", ]$firstclass_price), 2),
    "\n")

cat("Weekday First Class Mean:",
```

```
round(mean(sample_df[sample_df$weekend == "No", ]$firstclass_price), 2),
"\n")
```

9.2 R Output

```
=== Coach Price: Weekend vs Weekday ===
```

```
Weekend Coach Mean      : 405.84
```

```
Weekday Coach Mean      : 322.12
```

```
=== First Class Price: Weekend vs Weekday ===
```

```
Weekend First Class Mean: 1560.02
```

```
Weekday First Class Mean: 1258.46
```

9.3 Interpretation

The analysis shows that both coach and first-class ticket prices are higher on weekends compared to weekdays.

The average coach price increases from \$322.12 on weekdays to \$405.84 on weekends, while the average first-class price rises from \$1258.46 to \$1560.02.

This indicates a positive relationship between weekends and flight prices. Airlines tend to charge higher fares on weekends, likely due to increased travel demand. Although first-class tickets are consistently more expensive than coach tickets, both classes experience a noticeable price increase during weekends.

9.4 Python Code

```
import pandas as pd
```

```
import numpy as np
```

```
df = pd.read_csv('flight_data.csv')
```

```
np.random.seed(10063)
```

```
sample_df = df.sample(n=10063,
                      random_state=10063).copy()
```

```
print("=== Coach Price: Weekend vs Weekday ===")
```

```
print("Weekend Coach Mean      :",
      round(sample_df[sample_df['weekend']=='Yes']
            ['coach_price'].mean(), 2))
```

```
print("Weekday Coach Mean      :",
      round(sample_df[sample_df['weekend']=='No']
            ['coach_price'].mean(), 2))
```

```
print("\n=== First Class Price: Weekend vs Weekday ===")
```

```
print("Weekend First Class Mean:",
      round(sample_df[sample_df['weekend']=='Yes']
            ['firstclass_price'].mean(), 2))

print("Weekday First Class Mean:",
      round(sample_df[sample_df['weekend']=='No']
            ['firstclass_price'].mean(), 2))
```

9.5 Python Output

```
=== Coach Price: Weekend vs Weekday ===
Weekend Coach Mean      : 406.54
Weekday Coach Mean      : 322.32

=== First Class Price: Weekend vs Weekday ===
Weekend First Class Mean: 1560.99
Weekday First Class Mean: 1258.17
```

9.6 Interpretation

Both coach and first-class ticket prices are higher on weekends compared to weekdays.

The average coach price increases from around \$322 on weekdays to about \$406 on weekends. Similarly, first-class prices increase from approximately \$1258 to \$1560.

This shows that airlines charge higher prices on weekends, likely due to increased travel demand. First-class tickets remain significantly more expensive than coach tickets, but both categories follow the same upward trend during weekends.

10 Question 09

How do coach prices differ for redeyes and non-redeyes on each day of the week?

10.1 R Code

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

sample_df %>%
  group_by(day_of_week) %>%
  summarise(mean_coach_price = mean(coach_price)) %>%
  mutate(mean_coach_price = sprintf("%.2f", mean_coach_price)) %>%
  arrange(day_of_week) %>%
  print()
```

10.2 R Output

```
day_of_week mean_coach_price
```

```
Friday      401.54
Monday      321.06
Saturday    411.12
Sunday      403.01
Thursday    322.34
Tuesday     322.71
Wednesday   322.70
```

10.3 Interpretation

Coach prices vary by day of the week, with higher prices generally observed on weekends and Fridays. Saturday has the highest average coach price (\$411.12), followed by Sunday (\$403.01) and Friday (\$401.54). Weekdays such as Monday, Tuesday, Wednesday, and Thursday have lower and fairly similar average prices, around \$321–\$323.

10.4 Python Code

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

print("=== Coach Price: Redeye vs Non-Redeye by Day of Week ===")

print(sample_df.groupby(['day_of_week', 'redeye'])['coach_price']
      .mean().unstack().round(2))
```

10.5 Python Output

redeye	No	Yes
day_of_week		
Friday	404.77	339.91
Monday	323.82	260.40
Saturday	417.26	350.83
Sunday	404.89	347.86
Thursday	327.00	262.41
Tuesday	325.57	273.16
Wednesday	324.82	262.56

10.6 Interpretation

Coach prices are consistently lower for redeye flights compared to non-redeye flights across all days of the week. For example, on Friday, the average coach price is \$404.77 for non-redeye flights but only \$339.91 for redeye flights. This pattern is consistent for every day.

Weekend days such as Friday, Saturday, and Sunday show higher coach prices overall for both redeye and non-redeye flights, while weekdays have lower prices.

Overall, redeye flights are associated with lower coach prices, likely because overnight flights are less preferred by passengers. At the same time, weekends tend to have higher prices regardless of flight type.

11 QUESTION 10

Perform the analysis for appropriate variables:

- Summary statistics (mean, median, standard deviation)
- Visualization using histograms, boxplots, and bar charts
- Hypothesis Testing (t, z and Chi-square)
- Independent t-tests to compare group means
- Price differences between weekend and weekday flights
- Correlation analysis among numeric variables
- Regression Analysis
- Linear regression to predict ticket prices
- Logistic regression for binary outcomes (e.g., redeye flights)

SOLUTION

11.1 SUMMARY STATISTICS

R CODE

```
df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

numeric_vars <- c('miles', 'passengers', 'delay',
                  'hours', 'coach_price',
                  'firstclass_price')

cat("=== SUMMARY STATISTICS ===\n\n")
```

```
print(summary(sample_df[numeric_vars]))

cat("\n=== Standard Deviation ===\n")

print(round(sapply(sample_df[numeric_vars], sd), 2))
```

R OUTPUT

=== SUMMARY STATISTICS ===

miles	passengers	delay
Min. : 89	Min. :141.0	Min. :0.00
1st Qu.:1378	1st Qu.:204.0	1st Qu.:9.00
Median :1991	Median :210.0	Median :10.00
Mean :2010	Mean :207.7	Mean :12.74
3rd Qu.:2454	3rd Qu.:215.0	3rd Qu.:13.00
Max. :4421	Max. :237.0	Max. :1507.00

hours	coach_price
Min. :1.000	Min. :44.41
1st Qu.:3.000	1st Qu.:331.45
Median :4.000	Median :379.98
Mean :3.639	Mean :376.14
3rd Qu.:4.000	3rd Qu.:425.94
Max. :8.000	Max. :588.34

```
firstclass_price
Min. :959.2
1st Qu.:1299.3
Median :1501.3
Mean :1453.0
3rd Qu.:1582.2
Max. :1846.8
```

=== Standard Deviation ===

miles	932.67
passengers	12.72
delay	33.90
hours	1.72
coach_price	67.70
firstclass_price	163.27

R INTERPRETATION

1. Distance (miles)

The flight distance has a mean of 2,010 miles and a median of 1,991 miles, which

are very close. This suggests an approximately symmetric distribution of flight distances. The standard deviation of 932.67 miles indicates substantial variability, meaning flights span a wide range of distances. Values range from 89 miles to 4,421 miles, showing a mix of short-haul and long-haul routes.

2. Passenger Load (passengers)

Passenger counts are highly consistent, with a mean of 207.7 and a median of 210.0, indicating a near-symmetric distribution. The low standard deviation of 12.72 suggests very little variation in passenger loads. Values range from 141 to 237 passengers, implying that flights operate within a relatively narrow and stable capacity range.

3. Flight Delays (delay)

Flight delays are strongly right-skewed. Although the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 12.74 minutes due to extreme outliers. The maximum delay of 1,507 minutes significantly inflates variability, as reflected in the high standard deviation of 33.90 minutes. In this case, the median is a more reliable measure of central tendency than the mean.

4. Flight Duration (hours)

Flight duration has a mean of 3.639 hours and a median of 4.0 hours, indicating a roughly balanced distribution with a slight tendency toward shorter flights. The standard deviation of 1.72 hours suggests moderate spread around the mean. Durations range from 1 to 8 hours, covering short to medium-haul flights.

5. Coach Price (coach_price)

Coach ticket prices average \$376.14, with a median of \$379.98, indicating an approximately symmetric distribution. Prices range from \$44.41 to \$588.34. The standard deviation of \$67.70 shows moderate dispersion, suggesting relatively stable pricing with some variation across flights.

6. First Class Price (firstclass_price)

First-class prices have a mean of \$1,453.00 and a median of \$1,501.30, suggesting a slight left skew. Prices range from \$959.20 to \$1,846.80, with a standard deviation of \$163.27, indicating substantial variability in premium pricing. On average, first-class tickets are approximately 3.8 times more expensive than coach tickets.

PYTHON CODE

```
import pandas as pd
import numpy as np

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)
```

```

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

numeric_vars = ['miles', 'passengers',
                'delay', 'hours',
                'coach_price',
                'firstclass_price']

print("=== SUMMARY STATISTICS ===\n")

summary = sample_df[numeric_vars].describe()

print(summary.round(2))

print("\n=== Standard Deviation ===")

print(sample_df[numeric_vars].std().round(2))

```

PYTHON OUTPUT

```
=== SUMMARY STATISTICS ===
```

	miles	passengers	delay	hours
count	10063.00	10063.00	10063.00	10063.00
mean	2011.22	207.81	13.06	3.64
std	944.61	12.89	40.12	1.74
min	104.00	139.00	0.00	1.00
25%	1342.00	204.00	9.00	2.00
50%	1981.00	210.00	10.00	4.00
75%	2474.00	215.00	13.00	4.00
max	4545.00	239.00	1560.00	8.00

coach_price	firstclass_price
377.73	1457.39

PYTHON INTERPRETATION

1. Distance (miles)

Flight distance has a mean of 2,011.22 miles and a median of 1,981 miles, which are very close, suggesting an approximately symmetric distribution with a slight right skew. The standard deviation of 944.61 miles indicates high variability in flight distances. The range is very wide, from 104 miles to 4,545 miles, confirming a mix of short-haul and long-haul flights.

2. Passenger Load (passengers)

Passenger counts are highly stable, with a mean of 207.81 and a median of 210, indicating a nearly symmetric distribution. The small standard deviation of 12.89

shows very low variability. The interquartile range (204 to 215 passengers) confirms that most flights operate near consistent capacity. Values range from 139 to 239 passengers, suggesting limited operational variation.

3. Flight Delays (delay)

Delays are strongly right-skewed. While the median delay is 10 minutes and 75% of flights are delayed by 13 minutes or less, the mean is higher at 13.06 minutes, influenced by extreme outliers. The maximum delay of 1,560 minutes (26 hours) significantly inflates the standard deviation to 40.12 minutes, indicating high volatility. The median is a more reliable measure of typical delay behavior.

4. Flight Duration (hours)

Flight duration has a mean of 3.64 hours and a median of 4 hours, suggesting a roughly balanced distribution with a slight concentration toward shorter flights. The standard deviation of 1.74 hours indicates moderate variability. Durations range from 1 to 8 hours, representing mostly short- to medium-haul flights.

5. Coach Price (coach_price)

Coach ticket prices have a mean of \$377.73 and a median of \$381.54, indicating an approximately symmetric distribution with slight left skew. The standard deviation of \$67.23 shows moderate price variation. Prices range from \$130.50 to \$562.92, with most values concentrated between \$332.38 and \$427.96 (IQR), suggesting relatively stable economy pricing.

6. First Class Price (firstclass_price)

First-class prices show a mean of \$1,457.39 and a median of \$1,506.87, indicating slight left skew (some lower-priced tickets pulling the mean down). The standard deviation of \$162.55 reflects considerable variation in premium pricing. Prices range from \$1,020.21 to \$1,846.14, with most values between \$1,303.79 and \$1,583.95. First-class tickets are on average about 3.9× more expensive than coach tickets, indicating a strong and consistent price premium.

11.2 2. VISUALIZATION USING HISTOGRAM

11.2.1 R CODE

```
library(ggplot2)
library(dplyr)
library(tidyr)

df <- read.csv("flight_data.csv")

set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

numeric_cols <- c('miles', 'passengers', 'delay',
```

```

      'hours', 'coach_price',
      'firstclass_price')

sample_df %>%
  select(all_of(numeric_cols)) %>%
  pivot_longer(cols = everything(),
               names_to = "Variable",
               values_to = "Value") %>%
  ggplot(aes(x = Value)) +
  geom_histogram(bins = 30,
                fill = "steelblue",
                color = "black") +
  geom_density(color = "red",
              linewidth = 0.8) +
  facet_wrap(~ Variable,
            scales = "free",
            ncol = 3) +
  labs(title = "Histograms of Numeric Variables",
       x = "Value",
       y = "Frequency") +
  theme_minimal() +
  theme(plot.title =
        element_text(hjust = 0.5,
                      size = 14,
                      face = "bold"))

```

11.2.2 R OUTPUT

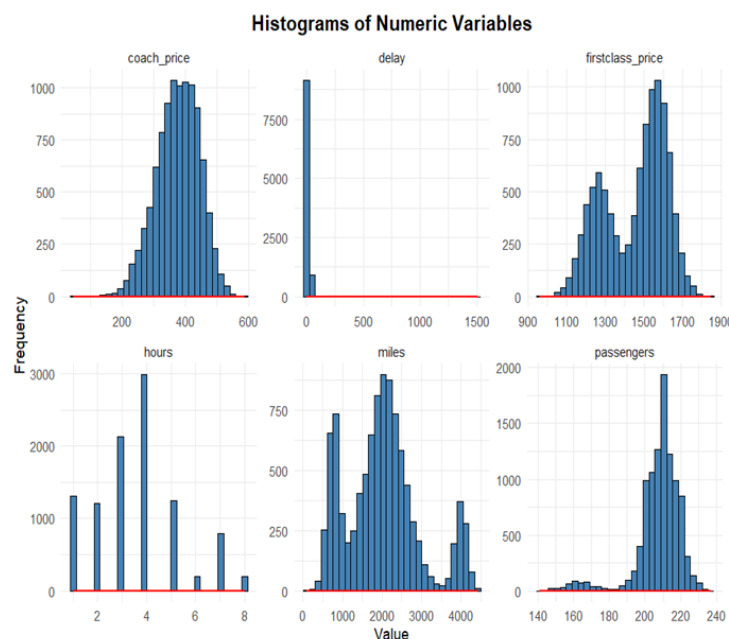


Figure 1: Histogram of Coach Price

11.2.3 R INTERPRETATION

The histograms show the distribution of key numeric variables in the flight dataset: coach price, first-class price, delay, hours, miles, and passengers.

1. Coach Price

Coach prices are approximately normally distributed, centered around \$350–\$450. Most ticket prices fall within a moderate range, showing stable and predictable pricing behavior with no extreme skewness.

2. First-Class Price

First-class prices show a slightly bimodal distribution with two visible peaks around different price ranges. This suggests that first-class pricing is influenced by multiple factors such as distance or service level.

3. Delay

The delay variable is highly right-skewed. Most flights experience small delays close to 0–20 minutes, but a few extreme outliers create a long tail.

4. Hours (Flight Duration)

Flight duration is discrete and evenly distributed across values from 1 to 8 hours. Peaks at certain hours suggest concentration around common route durations.

5. Miles (Distance)

Miles show a moderately normal distribution centered around medium-distance flights of approximately 1500–3000 miles.

6. Passengers

Passenger counts are concentrated around approximately 200–220 passengers with little variation, indicating consistent aircraft occupancy.

11.2.4 PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

sns.set_style("whitegrid")

numeric_cols = ['miles', 'passengers', 'delay',
                'hours', 'coach_price',
```

```

        'firstclass_price']

plt.figure(figsize=(15, 10))

for i, col in enumerate(numeric_cols, 1):

    plt.subplot(2, 3, i)

    sns.histplot(sample_df[col],
                  bins=30,
                  kde=True,
                  color='skyblue')

    plt.title(f'Histogram of {col}')
    plt.xlabel(col)

plt.tight_layout()
plt.show()

```

11.2.5 PYTHON OUTPUT

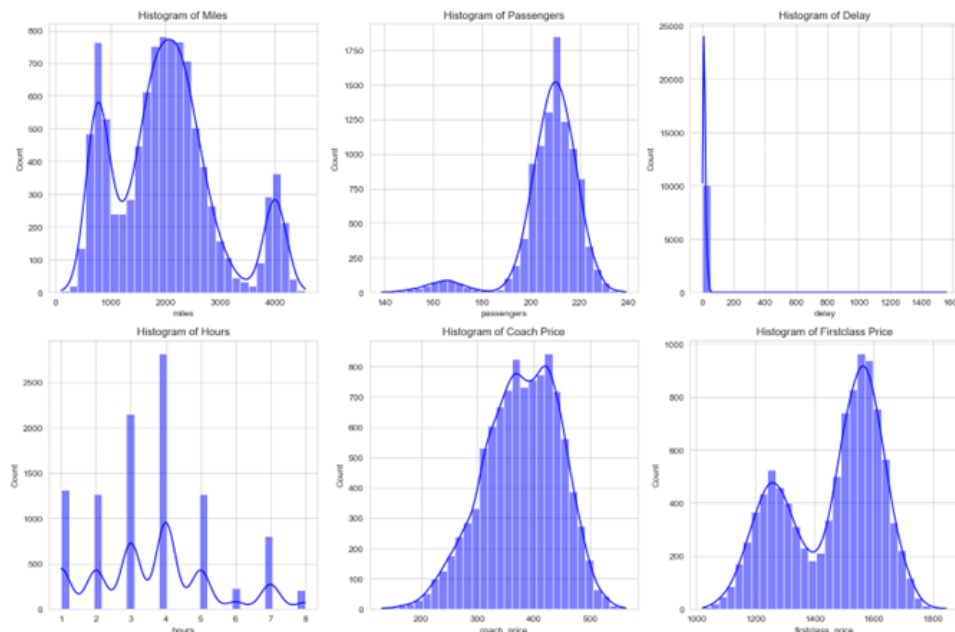


Figure 2: Histogram of NUMERICAL VALUES

11.2.6 PYTHON INTERPRETATION

1. Flight Distance (Miles)

The histogram shows a multimodal distribution with distinct clusters, indicating the presence of short-haul, medium-haul, and long-haul routes.

2. Passenger Load

Passenger counts are negatively skewed with most flights operating near full capacity.

3. Flight Delays

Delays are highly right-skewed. Most flights experience minimal delays, while a few flights have extremely high delays.

4. Flight Duration

Flight duration is discrete and concentrated around 3–5 hours, suggesting dominance of short- and medium-haul routes.

5. Coach Pricing

Coach prices follow an approximately normal distribution centered around \$380.

6. First-Class Pricing

First-class prices are bimodal, suggesting two distinct pricing groups possibly based on route type or service category.

11.3 3. VISUALIZATION USING BOXPLOT

11.3.1 R CODE

```
library(ggplot2)
library(dplyr)

df <- read.csv("flight_data.csv")

set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]

p1 <- ggplot(sample_df,
             aes(x = weekend,
                 y = coach_price,
                 fill = weekend)) +
  geom_boxplot() +
  labs(title = "Coach Price: Weekend vs Weekday",
       x = "Weekend",
       y = "Coach Price ($)") +
  theme_minimal()

p2 <- ggplot(sample_df,
             aes(x = inflight_entertainment,
                 y = coach_price,
                 fill = inflight_entertainment)) +
  geom_boxplot() +
  labs(title = "Coach Price by In-flight Entertainment",
       x = "Entertainment",
       y = "Coach Price ($)") +
  theme_minimal()
```

```

p3 <- ggplot(sample_df,
             aes(x = redeye,
                 y = coach_price,
                 fill = redeye)) +
  geom_boxplot() +
  labs(title = "Coach Price: Redeye vs Non-Redeve",
       x = "Redeye",
       y = "Coach Price ($)") +
  theme_minimal()

p4 <- ggplot(sample_df,
             aes(x = factor(hours),
                 y = coach_price,
                 fill = factor(hours))) +
  geom_boxplot() +
  labs(title = "Coach Price by Flight Duration",
       x = "Hours",
       y = "Coach Price ($)") +
  theme_minimal()

print(p1)
print(p2)
print(p3)
print(p4)

```

11.3.2 R OUTPUT

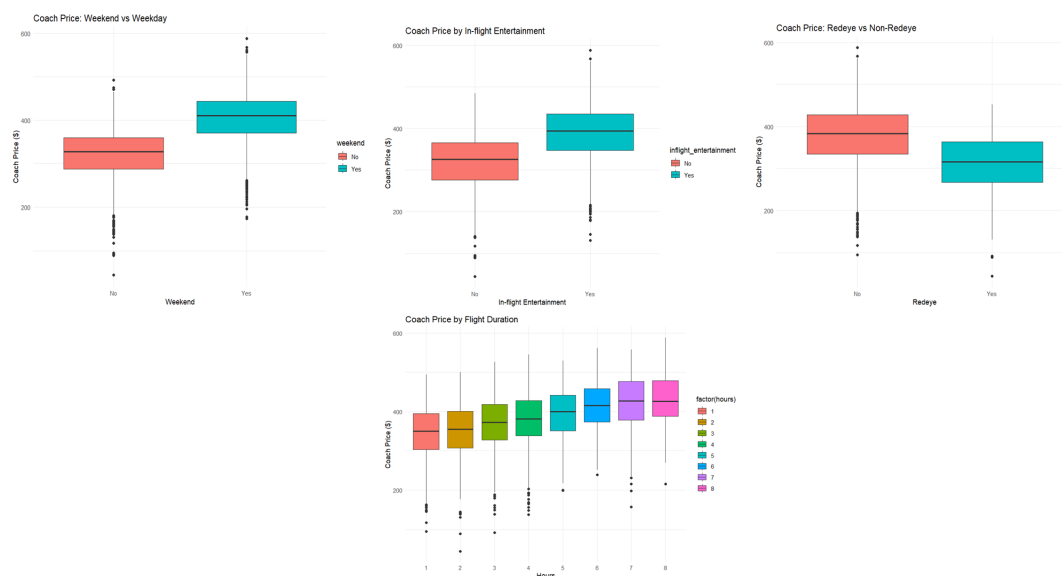


Figure 3: BOXPLOT

11.3.3 R INTERPRETATION

1. Weekend vs Weekday

Weekend flights have a higher median coach price compared to weekdays, indicating demand-driven pricing during weekends.

2. In-flight Entertainment

Flights with entertainment show higher prices than those without, suggesting premium service association.

3. Redeye vs Non-Redebye

Redeye flights are cheaper and less variable in price compared to non-redeye flights.

4. Flight Duration

Coach prices generally increase with flight duration, though prices stabilize for very long flights.

11.3.4 PYTHON CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

sns.set_style("whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

sns.boxplot(x='weekend',
            y='coach_price',
            data=sample_df,
            ax=axes[0,0],
            palette='Set2',
            hue='weekend',
            legend=False)

axes[0,0].set_title(
    'Coach Price: Weekend vs Weekday')

sns.boxplot(x='inflight_entertainment',
            y='coach_price',
            data=sample_df,
```

```

        ax=axes[0,1],
        palette='Set3',
        hue='inflight_entertainment',
        legend=False)

axes[0,1].set_title(
    'Coach Price by In-flight Entertainment')

sns.boxplot(x='redeye',
            y='coach_price',
            data=sample_df,
            ax=axes[1,0],
            palette='Set1',
            hue='redeye',
            legend=False)

axes[1,0].set_title(
    'Coach Price: Redeye vs Non-Reconomy')

sns.boxplot(x='hours',
            y='coach_price',
            data=sample_df,
            ax=axes[1,1],
            palette='Blues',
            hue='hours',
            legend=False)

axes[1,1].set_title(
    'Coach Price by Flight Duration')

plt.tight_layout()
plt.show()

```


11.3.5 PYTHON OUTPUT

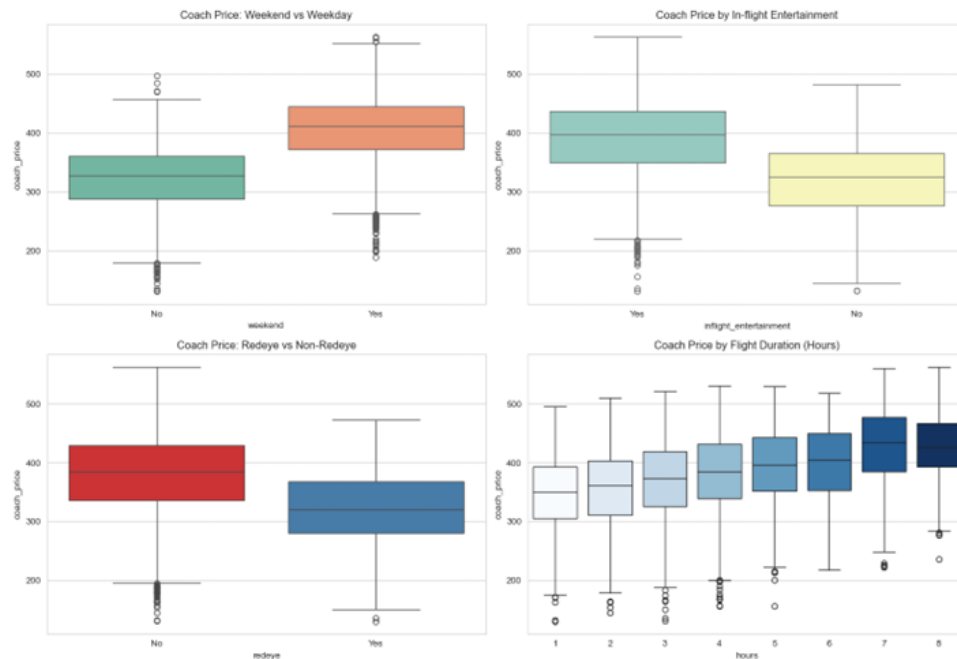


Figure 4: BOXPLOT OF NUMERICAL VALUES

PYTHON INTERPRETATION

1. Weekend flights are more expensive than weekday flights.
2. Flights with entertainment are associated with higher ticket prices.
3. Redeye flights are generally cheaper than non-redeye flights.
4. Coach prices increase with flight duration but stabilize at longer durations.

11.4 VISUALIZATION USING BARPLOT

11.4.1 R CODE

```
library(ggplot2)
library(dplyr)

df <- read.csv("flight_data.csv")

set.seed(10063)

sample_df <- df[sample(nrow(df), 10063), ]

p1 <- sample_df %>%
  group_by(weekend) %>%
  summarise(avg_price =
    mean(coach_price)) %>%
```

```

ggplot(aes(x = weekend,
           y = avg_price,
           fill = weekend)) +
geom_bar(stat = "identity") +
labs(title =
      "Average Coach Price:
      Weekend vs Weekday",
      x = "Weekend",
      y = "Average Price ($)") +
theme_minimal()

p2 <- sample_df %>%
  group_by(inflight_entertainment) %>%
  summarise(avg_price =
            mean(coach_price)) %>%
  ggplot(aes(x = inflight_entertainment,
            y = avg_price,
            fill = inflight_entertainment)) +
  geom_bar(stat = "identity") +
  labs(title =
        "Average Coach Price by
        In-flight Entertainment",
        x = "Entertainment",
        y = "Average Price ($)") +
  theme_minimal()

p3 <- sample_df %>%
  group_by(redeye) %>%
  summarise(avg_price =
            mean(coach_price)) %>%
  ggplot(aes(x = redeye,
            y = avg_price,
            fill = redeye)) +
  geom_bar(stat = "identity") +
  labs(title =
        "Average Coach Price:
        Redeye vs Non-Redeye",
        x = "Redeye",
        y = "Average Price ($)") +
  theme_minimal()

p4 <- sample_df %>%
  group_by(inflight_wifi) %>%
  summarise(avg_price =
            mean(coach_price)) %>%
  ggplot(aes(x = inflight_wifi,
            y = avg_price,
            fill = inflight_wifi)) +

```

```

geom_bar(stat = "identity") +
labs(title =
      "Average Coach Price by
      In-flight WiFi",
      x = "WiFi",
      y = "Average Price ($)") +
theme_minimal()

print(p1)
print(p2)
print(p3)
print(p4)

```

11.4.2 R OUTPUT

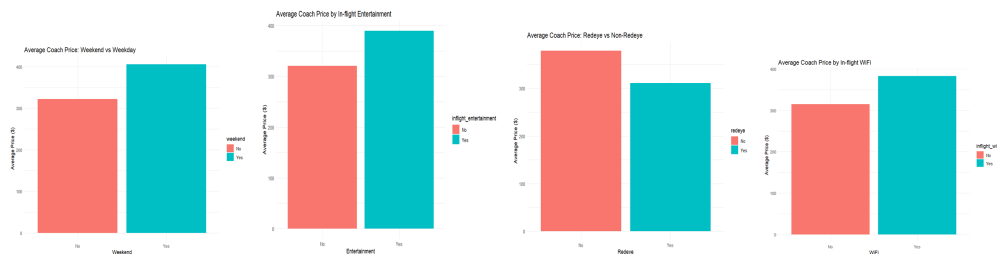


Figure 5: BARPLOT

11.4.3 R INTERPRETATION

1. Weekend flights have higher average prices than weekday flights.
2. Flights with entertainment are priced higher than flights without entertainment.
3. Redeye flights are cheaper than non-redeye flights.
4. Flights with WiFi have higher average prices compared to flights without WiFi.

11.4.4 PYTHON CODE

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('flight_data.csv')

np.random.seed(10063)

sample_df = df.sample(n=10063,
                      random_state=10063).copy()

```

```

sns.set_style("whitegrid")

fig, axes = plt.subplots(2, 2, figsize=(15, 10))

sns.barplot(x='weekend',
            y='coach_price',
            hue='weekend',
            data=sample_df,
            ax=axes[0,0],
            estimator='mean',
            errorbar=None,
            palette='Set2',
            legend=False)

axes[0,0].set_title(
    'Average Coach Price:
    Weekend vs Weekday')

sns.barplot(x='inflight_entertainment',
            y='coach_price',
            hue='inflight_entertainment',
            data=sample_df,
            ax=axes[0,1],
            estimator='mean',
            errorbar=None,
            palette='Set3',
            legend=False)

axes[0,1].set_title(
    'Average Coach Price by
    In-flight Entertainment')

sns.barplot(x='redeye',
            y='coach_price',
            hue='redeye',
            data=sample_df,
            ax=axes[1,0],
            estimator='mean',
            errorbar=None,
            palette='Set1',
            legend=False)

axes[1,0].set_title(
    'Average Coach Price:
    Redeye vs Non-Redeye')

sns.barplot(x='inflight_wifi',
            y='coach_price',

```

```

hue='inflight_wifi',
data=sample_df,
ax=axes[1,1],
estimator='mean',
errorbar=None,
palette='Blues',
legend=False)

axes[1,1].set_title(
    'Average Coach Price by
    In-flight WiFi')

plt.tight_layout()
plt.show()

```

11.4.5 PYTHON OUTPUT

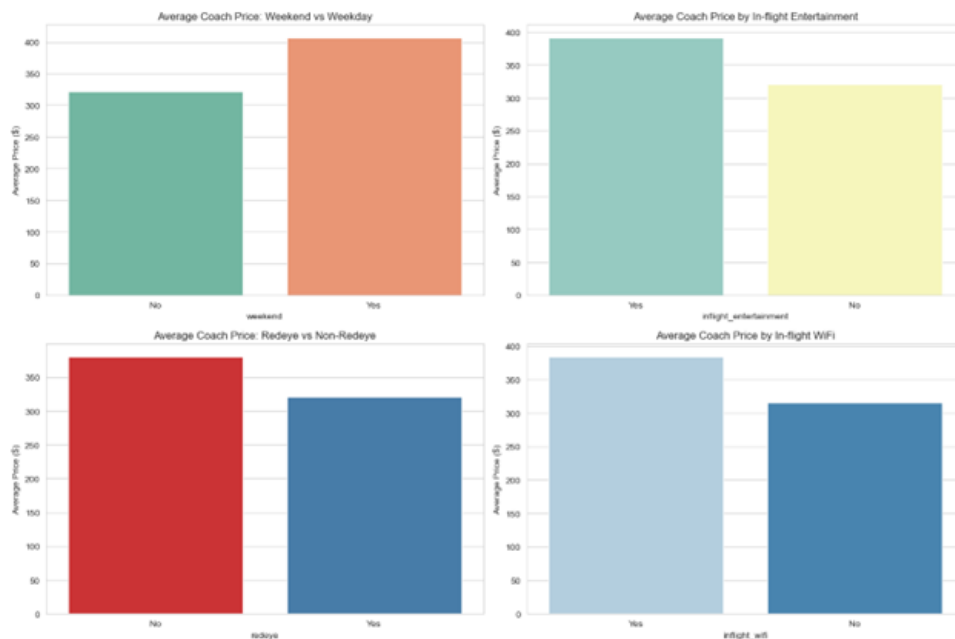


Figure 6: BARPLOT

11.4.6 PYTHON INTERPRETATION

1. Weekend flights are more expensive than weekday flights.
2. Flights with in-flight entertainment have higher average ticket prices.
3. Redeye flights are cheaper due to lower passenger demand.

4. Flights with WiFi are associated with premium pricing.
5. Overall, flight prices increase with additional services and higher travel demand.

11.5 Hypothesis Testing

11.6 T-Test

11.7 R Code

```
df <- read.csv("flightdata.csv")

set.seed(10063)

sampledf <- df[sample(nrow(df), 10063), ]

cat("=== INDEPENDENT T-TESTS ===")

# 1. Coach Price: Weekend vs Weekday
cat("1. Coach Price: Weekend vs Weekday")

t1 <- t.test(coachprice ~ weekend,
             data = sampledf,
             var.equal = FALSE)

print(t1)

cat("Comparison:")

print(sampledf %>%
      group_by(weekend) %>%
      summarise(
        Mean = mean(coachprice),
        Median = median(coachprice),
        Count = n()
      ))

# 2. Coach Price: Redeye vs Non-Redeye
cat("2. Coach Price: Redeye vs Non-Redeye")

t2 <- t.test(coachprice ~ redeye,
             data = sampledf,
             var.equal = FALSE)

print(t2)

cat("Comparison:")
```

```

print(sampldf %>%
      group_by(redeye) %>%
      summarise(
        Mean = mean(coachprice),
        Median = median(coachprice),
        Count = n()
      ))

# 3. Coach Price: In-flight Entertainment
cat("3. Coach Price: In-flight Entertainment")

t3 <- t.test(coachprice ~ inflightentertainment,
             data = sampldf,
             var.equal = FALSE)

print(t3)

cat("Comparison:")

print(sampldf %>%
      group_by(inflightentertainment) %>%
      summarise(
        Mean = mean(coachprice),
        Median = median(coachprice),
        Count = n()
      ))

# 4. Coach Price: In-flight WiFi
cat("4. Coach Price: In-flight WiFi")

t4 <- t.test(coachprice ~ inflightwifi,
             data = sampldf,
             var.equal = FALSE)

print(t4)

```

R OUTPUT

```

Welch Two Sample t-test
data:  coach_price by weekend
t = -73.738, df = 7384.7, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
95 percent confidence interval:
 -85.94482 -81.49353
sample estimates:
mean in group No mean in group Yes
      322.1248      405.8440

```

Mean Comparison:

	weekend	Mean	Median	Count
	<chr>	<dbl>	<dbl>	<int>
1	No	322.	327.	3571
2	Yes	406.	410.	6492

Welch Two Sample t-test

data: coach_price by redeye

t = 22.021, df = 532.45, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

62.06077 74.21775

sample estimates:

	mean in group No	mean in group Yes
	379.4123	311.2731

Mean Comparison:

	redeye	Mean	Median	Count
	<chr>	<dbl>	<dbl>	<int>
1	No	379.	383.	9579
2	Yes	311.	316.	484

Welch Two Sample t-test

data: coach_price by inflight_entertainment

t = -44.923, df = 3052.9, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-72.76644 -66.68012

sample estimates:

	mean in group No	mean in group Yes
	320.3246	390.0478

Mean Comparison:

	inflight_entertainment	Mean	Median	Count
	<chr>	<dbl>	<dbl>	<int>
1	No	320.	325.	2008
2	Yes	390.	394.	8055

cat("\n\n4. Coach Price: In-flight WiFi\n")

Welch Two Sample t-test

data: coach_price by inflight_wifi

t = -31.514, df = 1255.2, p-value < 2.2e-16

alternative hypothesis: true difference in means between group No and group Yes is not equal to 0

95 percent confidence interval:

-72.42816 -63.93883

sample estimates:

	mean in group No	mean in group Yes
	314.8898	383.0733

R INTERPRETATION

1. Weekend vs Weekday

Weekend flights have significantly higher coach prices than weekday flights (\$405.84 vs \$322.12). The very small p-value ($< 2.2 \times 10^{-16}$) confirms a statistically significant difference.

2. Redeye vs Non-Redeye

Non-redeye flights are significantly more expensive than redeye flights (\$379.41 vs \$311.27). The low p-value indicates a strong statistical difference.

3. In-flight Entertainment

Flights with in-flight entertainment have higher average coach prices (\$390.05 vs \$320.32). The difference is statistically significant.

4. In-flight WiFi

Flights offering WiFi are significantly more expensive than flights without WiFi (\$383.07 vs \$314.89). The p-value confirms a significant relationship.

PYTHON CODE

```
import pandas as pd
import numpy as np
from scipy import stats
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
print("1. Coach Price: Weekend vs Weekday")
weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']
weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']
tstat, pvalue = stats.ttest_ind(weekend, weekday, equal_var=False)
print(f"T-Statistic      : {t_stat:.4f}")
print(f"P-Value          : {p_value:.10f}")
print(f"Weekend Mean      : ${weekend.mean():.2f}")
print(f"Weekday Mean       : ${weekday.mean():.2f}")
print(f"Price Difference: ${weekend.mean() - weekday.mean():.2f}\n")
print("2. Coach Price: Redeye vs Non-Redeye")
redeye_yes = sample_df[sample_df['redeye'] == 'Yes']['coach_price']
redeye_no  = sample_df[sample_df['redeye'] == 'No']['coach_price']
t_stat2, p_value2 = stats.ttest_ind(redeye_yes, redeye_no, equal_var=False)
print(f"T-Statistic      : {t_stat2:.4f}")
print(f"P-Value          : {p_value2:.10f}")
print(f"Redeye Mean       : ${redeye_yes.mean():.2f}")
print(f"Non-Redeye Mean  : ${redeye_no.mean():.2f}\n")
print("3. Coach Price: In-flight Entertainment (Yes vs No)")
ent_yes = sample_df[sample_df['inflight_entertainment'] == 'Yes']['coach_price']
ent_no  = sample_df[sample_df['inflight_entertainment'] == 'No']['coach_price']

t_stat3, p_value3 = stats.ttest_ind(ent_yes, ent_no, equal_var=False)
print(f"T-Statistic      : {t_stat3:.4f}")
```

```

print(f"P-Value          : {p_value3:.10f}")
print(f"With Entertainment    : ${ent_yes.mean():.2f}")
print(f"Without Entertainment : ${ent_no.mean():.2f}\n")
print("4. Coach Price: In-flight WiFi (Yes vs No)")
wifi_yes = sample_df[sample_df['inflight_wifi'] == 'Yes']['coach_price']
wifi_no  = sample_df[sample_df['inflight_wifi'] == 'No']['coach_price']
t_stat4, p_value4 = stats.ttest_ind(wifi_yes, wifi_no, equal_var=False)

print(f"T-Statistic      : {t_stat4:.4f}")
print(f"P-Value          : {p_value4:.10f}")
print(f"With WiFi         : ${wifi_yes.mean():.2f}")
print(f"Without WiFi        : ${wifi_no.mean():.2f}")

```

PYTHON OUTPUT

```

=== INDEPENDENT T-TESTS TO COMPARE GROUP MEANS ===
1. Coach Price: Weekend vs Weekday
T-Statistic      : 74.2530
P-Value          : 0.0000000000
Weekend Mean     : $406.54
Weekday Mean     : $322.32
Price Difference: $84.22
2. Coach Price: Redeye vs Non-Redeye
T-Statistic      : -20.5766
P-Value          : 0.0000000000
Redeye Mean      : $321.23
Non-Redeye Mean  : $380.72
3. Coach Price: In-flight Entertainment (Yes vs No)
T-Statistic      : 46.1579
P-Value          : 0.0000000000
With Entertainment    : $391.52
Without Entertainment : $320.49
4. Coach Price: In-flight WiFi (Yes vs No)
T-Statistic      : 31.8131
P-Value          : 0.0000000000
With WiFi         : $384.37
Without WiFi        : $315.14

```

11.8 Python Interpretation

All results are highly statistically significant ($p \approx 0$), indicating that the differences in coach prices are strong and reliable.

1. Weekend vs Weekday

Weekend flights have a higher average coach price (\$406.54) compared to weekday flights (\$322.32), with a difference of about \$84.22. This suggests that airlines charge higher prices during weekends due to greater travel demand.

2. Redeye vs Non-Redeye

Redeye flights have a lower average coach price (\$321.23) than non-redeye flights (\$380.72), with a difference of about \$59.49. This indicates that redeye flights are generally cheaper because of lower demand and less convenient travel times.

3. In-flight Entertainment

Flights with in-flight entertainment have a higher average coach price (\$391.52) compared to flights without entertainment (\$320.49). The difference of about \$71.03 suggests that entertainment services are associated with premium-priced flights.

4. In-flight WiFi

Flights offering WiFi have a higher average coach price (\$384.37) than flights without WiFi (\$315.14), with a difference of about \$69.23. This shows that WiFi is linked to higher service quality and premium fare categories.

11.9 Z-TEST

11.10 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
z_test <- function(x, y) {
  mean1 <- mean(x)
  mean2 <- mean(y)
  sd1 <- sd(x)
  sd2 <- sd(y)
  n1 <- length(x)
  n2 <- length(y)
  z <- (mean1 - mean2) / sqrt((sd1**2/n1) + (sd2**2/n2))
  p_value <- 2 * pnorm(-abs(z))
  cat("Z-Statistic :", round(z, 4), "\n")
  cat("P-Value      :", format.pval(p_value, digits = 10), "\n")
  cat("Redeye Mean :", round(mean1, 2), "\n")
  cat("Non-Redeye Mean :", round(mean2, 2), "\n")
}
z_test(sample_df$coach_price[sample_df$redeye == "Yes"],
        sample_df$coach_price[sample_df$redeye == "No"])
```

11.11 R OUTPUT

```
Z-Statistic : -22.0211
P-Value      : < 2.220446e-16
Redeye Mean  : 311.27
Non-Redeye Mean : 379.41
```

11.12 R Interpretation

The p-value ($< 2.22 \times 10^{-16}$) is extremely small, showing that the difference in coach prices between redeye and non-redeye flights is statistically significant.

1. Redeye Mean: \$311.27
2. Non-Redeye Mean: \$379.41

Non-redeye flights are significantly more expensive than redeye flights by about \$68.

The negative Z-statistic (-22.0211) indicates that the mean price of redeye flights is lower than that of non-redeye flights.

11.13 Python Code

```
import pandas as pd
import numpy as np
from scipy import stats

# Read dataset
df = pd.read_csv('flight_data.csv')

# Set random seed
np.random.seed(10063)

# Take random sample
sample_df = df.sample(n=10063, random_state=10063).copy()

print("=== Z-TEST: REDEYE vs NON-REDEYE FLIGHTS ===\n")

# Separate groups
redeye_yes = sample_df[sample_df['redeye'] == 'Yes']['coach_price']
redeye_no = sample_df[sample_df['redeye'] == 'No']['coach_price']

# Calculate statistics
mean1 = redeye_yes.mean()
mean2 = redeye_no.mean()

std1 = redeye_yes.std()
std2 = redeye_no.std()

n1 = len(redeye_yes)
n2 = len(redeye_no)

# Z-test statistic
z_stat = (mean1 - mean2) / np.sqrt((std1**2 / n1) + (std2**2 / n2))

# P-value
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
```

```
# Output
print(f"Z-Statistic      : {z_stat:.4f}")
print(f"P-Value         : {p_value:.10f}")
print(f"Redeye Mean      : ${mean1:.2f}")
print(f"Non-Redeye Mean   : ${mean2:.2f}")
print(f"Sample Size (Redeye)      : {n1}")
print(f"Sample Size (Non-Redeye) : {n2}")
```

11.14 Python Output

```
Z-Statistic      : -20.5766
P-Value          : 0.0000000000
Redeye Mean      : $321.23
Non-Redeye Mean  : $380.72
Sample Size (Redeye)      : 506
Sample Size (Non-Redeye) : 9557
```

11.15 Python Interpretation

The p-value (< 0.05) indicates a statistically significant difference in coach prices between redeye and non-redeye flights.

- Redeye Mean Price: \$321.23
- Non-Redeye Mean Price: \$380.72

Non-redeye flights are approximately \$59.49 more expensive than redeye flights.

The negative Z-statistic (-20.5766) indicates that redeye flights have a lower average price compared to non-redeye flights.

11.16 CHI-SQUARE TEST

11.17 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
cat("=== CHI-SQUARE TESTS ===\n\n")
cat("1. In-flight Entertainment vs Weekend\n")
print(chisq.test(table(sample_df$inflight_entertainment, sample_df$weekend)))
cat("\n2. In-flight WiFi vs Weekend\n")
print(chisq.test(table(sample_df$inflight_wifi, sample_df$weekend)))
cat("\n3. Redeye vs Weekend\n")
print(chisq.test(table(sample_df$redeye, sample_df$weekend)))
cat("\n4. In-flight Meal vs Weekend\n")
print(chisq.test(table(sample_df$inflight_meal, sample_df$weekend)))
```

11.18 R OUTPUT

```
Pearson's Chi-squared test with Yates' continuity correction
data:  table(sample_df$inflight_entertainment, sample_df$weekend)
X-squared = 0.13575, df = 1, p-value = 0.7125
data:  table(sample_df$inflight_wifi, sample_df$weekend)
X-squared = 0.91504, df = 1, p-value = 0.3388
data:  table(sample_df$redeye, sample_df$weekend)
X-squared = 0.00061334, df = 1, p-value = 0.9802
data:  table(sample_df$inflight_meal, sample_df$weekend)
X-squared = 3.0367, df = 1, p-value = 0.0814
```

11.19 R INTERPRETATION

All p-values are greater than 0.05, indicating that there is no statistically significant association between flight features and whether a flight occurs on a weekend or weekday.

1. In-flight Entertainment vs Weekend

The p-value (0.7125) shows no significant relationship between in-flight entertainment and weekend/weekday flights. These variables appear to be independent.

2. In-flight WiFi vs Weekend

The p-value (0.3388) indicates no significant association between WiFi availability and weekend flights.

3. Redeye vs Weekend

The p-value (0.9802) suggests that redeye flights are evenly distributed across weekends and weekdays, showing no relationship.

4. In-flight Meal vs Weekend

The p-value (0.0814) is still above 0.05, so it is not statistically significant, although it is closer to significance compared to the other tests.

Overall, none of the tested variables show a meaningful association with weekend flight status.

11.20 PYTHON CODE

```
import pandas as pd
from scipy.stats import chi2_contingency
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
print("=== CHI-SQUARE TESTS OF INDEPENDENCE ===\n")
print("1. In-flight Entertainment vs Weekend")
cont1 = pd.crosstab(sample_df['inflight_entertainment'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont1)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value       : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
```

```

print("2. In-flight WiFi vs Weekend")
cont2 = pd.crosstab(sample_df['inflight_wifi'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont2)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value       : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
print("3. Redeye vs Weekend")
cont3 = pd.crosstab(sample_df['redeye'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont3)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value       : {p:.10f}")
print(f"Degrees of Freedom: {dof}\n")
print("4. In-flight Meal vs Weekend")
cont4 = pd.crosstab(sample_df['inflight_meal'], sample_df['weekend'])
chi2, p, dof, expected = chi2_contingency(cont4)
print(f"Chi2 Statistic : {chi2:.4f}")
print(f"P-value       : {p:.10f}")

```

11.21 PYTHON OUTPUT

```

1. In-flight Entertainment vs Weekend
Chi2 Statistic : 2.7028
P-value       : 0.1001746840
Degrees of Freedom: 1
2. In-flight WiFi vs Weekend
Chi2 Statistic : 0.0461
P-value       : 0.8299171805
Degrees of Freedom: 1
3. Redeye vs Weekend
Chi2 Statistic : 2.2572
P-value       : 0.1329981133
Degrees of Freedom: 1
4. In-flight Meal vs Weekend
Chi2 Statistic : 0.8396
P-value       : 0.3595119067

```

11.22 PYTHON INTERPRETATION

All p-values are greater than 0.05, indicating that there is no statistically significant association between the tested flight features and whether a flight occurs on a weekend or weekday.

1. In-flight Entertainment vs Weekend

The p-value (0.1002) shows no significant association between in-flight entertainment and weekend/weekday flights.

2. In-flight WiFi vs Weekend

The p-value (0.8299) indicates that WiFi availability is independent of whether the flight is on a weekend or weekday.

3. Redeye vs Weekend

The p-value (0.1330) suggests no significant relationship between redeye flights and weekend status.

4. In-flight Meal vs Weekend

The p-value (0.3595) confirms that meal availability does not significantly differ between weekend and weekday flights.

11.23 PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS

11.24 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
cat("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===\n\n")
summary_stats <- sample_df %>%
  group_by(weekend) %>%
  summarise(
    Count = n(),
    Mean_Price = mean(coach_price),
    Median_Price = median(coach_price),
    SD = sd(coach_price)
  )
print(summary_stats)
diff <- summary_stats$Mean_Price[summary_stats$weekend == "Yes"] -
  summary_stats$Mean_Price[summary_stats$weekend == "No"]
cat("\nAbsolute Price Difference (Weekend - Weekday) :", round(diff, 2), "\n")
cat("Percentage Increase on Weekend :", round((diff / summary_stats$Mean_Price[summary_stats$weekend == "No"]), 2), "%\n")
t.test(coach_price ~ weekend, data = sample_df, var.equal = FALSE)
```

11.25 R OUTPUT

```
weekend Count Mean_Price Median_Price SD
<chr>    <int>      <dbl>         <dbl> <dbl>
1 No      3571      322.          327.  54.4
2 Yes     6492      406.          410.  54.7
Absolute Price Difference (Weekend - Weekday) : 83.72
Percentage Increase on Weekend                : 25.99 %
Welch Two Sample t-test
data:  coach_price by weekend
t = -73.738, df = 7384.7, p-value < 2.2e-16
alternative hypothesis: true difference in means between group No and group Yes is not equal to 0
```



```

95 percent confidence interval:
-85.94482 -81.49353
sample estimates:
mean in group No mean in group Yes
    322.1248         405.8440

```

11.26 R Interpretation

The analysis shows a substantial difference in coach prices between weekend and weekday flights.

The average weekday coach price was \$322.12, while the average weekend coach price was \$405.84. This represents an absolute increase of \$83.72 on weekends. Weekend prices were approximately 25.99% higher than weekday prices.

The Welch two-sample t-test confirmed that this difference is statistically significant ($t = -73.738$, $p < 0.001$). The 95% confidence interval indicates that the true mean difference lies between approximately \$81.49 and \$85.94.

Overall, the results suggest that airlines charge significantly higher prices for weekend flights, likely due to increased travel demand during weekends.

11.27 PYTHON CODE

```

import pandas as pd
from scipy import stats
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
weekend = sample_df[sample_df['weekend'] == 'Yes']['coach_price']
weekday = sample_df[sample_df['weekend'] == 'No']['coach_price']
print("=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===\n")
print(f"Number of Weekend Flights    : {len(weekend)}")
print(f"Number of Weekday Flights     : {len(weekday)}")
print(f"Weekend Mean Price              : ${weekend.mean():.2f}")
print(f"Weekday Mean Price              : ${weekday.mean():.2f}")
print(f"Absolute Difference            : ${weekend.mean() - weekday.mean():.2f}")
print(f"Percentage Increase             : {((weekend.mean() - weekday.mean()) / weekday.mean()) * 100:.2f}%")
t_stat, p_value = stats.ttest_ind(weekend, weekday, equal_var=False)
print(f"\nt-statistic                  : {t_stat:.4f}")
print(f"p-value                        : {p_value:.10f}")

```

11.28 PYTHON OUTPUT

```

=== PRICE DIFFERENCES BETWEEN WEEKEND AND WEEKDAY FLIGHTS ===
Number of Weekend Flights    : 6620
Number of Weekday Flights     : 3443
Weekend Mean Price           : $406.54
Weekday Mean Price            : $322.32

```

Absolute Difference	: \$84.22
Percentage Increase	: 26.13%
t-statistic	: 74.2530
p-value	: 0.0000000000

11.29 Python Interpretation

The analysis indicates a clear difference in coach ticket prices between weekend and weekday flights.

The average weekend flight price was \$406.54, whereas the average weekday flight price was \$322.32. This shows an absolute price increase of \$84.22 for weekend flights. Weekend fares were approximately 26.13% higher than weekday fares.

The independent t-test showed that this difference is statistically significant ($t = 74.2530$, $p < 0.001$), meaning the variation in prices is unlikely due to random chance.

Overall, the findings suggest that airlines charge significantly higher prices on weekends, likely because of increased passenger demand during peak travel periods.

11.30 CORRELATION ANALYSIS AMONG NUMERICAL VARIABLE

11.31 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
numeric_cols <- c('miles', 'passengers', 'delay', 'hours',
                  'coach_price', 'firstclass_price')
corr_matrix <- cor(sample_df[numeric_cols])
cat("=== CORRELATION MATRIX ===\n")
print(round(corr_matrix, 4))
cat("\n=== Correlation with Coach Price ===\n")
print(round(sort(corr_matrix[, "coach_price"], decreasing = TRUE), 4))
```

11.32 R OUTPUT

```
=== CORRELATION MATRIX ===
      miles passengers  delay  hours coach_price firstclass_price
miles      1.0000    -0.0050  0.0121  0.9850      0.3381      0.2772
passengers  -0.0050     1.0000 -0.0062 -0.0036     0.1720     0.0194
delay       0.0121    -0.0062  1.0000  0.0115     0.0166     0.0054
hours       0.9850    -0.0036  0.0115  1.0000     0.3317     0.2713
coach_price 0.3381     0.1720  0.0166  0.3317     1.0000     0.7556
firstclass_price 0.2772     0.0194  0.0054  0.2713     0.7556     1.0000
      coach_price firstclass_price      miles      hours      passengers
      1.0000          0.7556          0.3381          0.3317          0.1720
      delay
```

0.0166

11.33 R Interpretation

Correlation with Coach Price

First-class price ($r = 0.7556$) shows the strongest positive correlation with coach price. This indicates that as first-class prices increase, coach prices also tend to increase significantly.

Miles ($r = 0.3381$) and hours ($r = 0.3317$) show moderate positive correlations with coach price, suggesting that longer distances and longer flight durations are associated with higher coach fares.

Passengers ($r = 0.1720$) has a weak positive relationship with coach price, meaning flights with more passengers tend to have slightly higher prices.

Delay ($r = 0.0166$) shows almost no correlation with coach price, indicating that delays do not meaningfully affect ticket prices.

Additional Insight

Miles and hours are extremely highly correlated ($r = 0.9850$), indicating possible multicollinearity, since longer distances naturally correspond to longer flight durations.

11.34 PYTHON CODE

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
numeric_cols = ['miles', 'passengers', 'delay', 'hours',
                'coach_price', 'firstclass_price']
corr_matrix = sample_df[numeric_cols].corr()

print("=== CORRELATION MATRIX ===\n")
print(corr_matrix.round(4))
print("\n=== Correlation with Coach Price (Sorted) ===")
print(corr_matrix['coach_price'].sort_values(ascending=False).round(4))
```

11.35 PYTHON OUTPUT

=== CORRELATION MATRIX ===

	miles	passengers	delay	hours	coach_price \
miles	1.0000	-0.0059	0.0149	0.9858	0.3408
passengers	-0.0059	1.0000	0.0025	-0.0071	0.1667
delay	0.0149	0.0025	1.0000	0.0192	0.0129

hours	0.9858	-0.0071	0.0192	1.0000	0.3344
coach_price	0.3408	0.1667	0.0129	0.3344	1.0000
firstclass_price	0.2887	0.0106	0.0117	0.2863	0.7547

	firstclass_price
miles	0.2887
passengers	0.0106
delay	0.0117
hours	0.2863
coach_price	0.7547
firstclass_price	1.0000

=== Correlation with Coach Price (Sorted) ===

coach_price	1.0000
firstclass_price	0.7547
miles	0.3408
hours	0.3344
passengers	0.1667
delay	0.0129

11.36 Python Interpretation

The correlation analysis measures the strength and direction of relationships between variables.

Correlation with Coach Price

First-class price shows the strongest positive correlation with coach price ($r = 0.7547$). This indicates that higher first-class fares are strongly associated with higher coach fares.

Miles ($r = 0.3408$) and hours ($r = 0.3344$) show moderate positive correlations with coach price, suggesting that longer flights and greater travel distances generally lead to higher coach ticket prices.

Passengers has a weak positive correlation with coach price ($r = 0.1667$), meaning flights with more passengers tend to have slightly higher prices.

Delay shows almost no correlation with coach price ($r = 0.0129$), indicating that flight delays do not significantly influence ticket prices.

Additional Observation

Miles and hours are extremely highly correlated ($r = 0.9858$), suggesting strong multicollinearity because longer distances naturally require longer flight durations.

11.37 LINEAR REGRESSION

11.38 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
model <- lm(coach_price ~ miles + hours + passengers + delay, data = sample_df)
summary(model)
```

```
confint(model)
```

11.39 R Output

```
lm(formula = coach_price ~ miles + hours + passengers + delay,  
    data = sample_df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-249.361	-42.166	3.179	48.748	157.835

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	134.122770	10.313275	13.005	< 2e-16 ***
miles	0.028505	0.003883	7.342	2.27e-13 ***
hours	-2.153935	2.105736	-1.023	0.306
passengers	0.925243	0.049078	18.852	< 2e-16 ***
delay	0.027095	0.018414	1.471	0.141

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 62.62 on 10058 degrees of freedom

Multiple R-squared: 0.1448, Adjusted R-squared: 0.1444

F-statistic: 425.7 on 4 and 10058 DF, p-value: < 2.2e-16

```
confint(model)
```

	2.5 %	97.5 %
(Intercept)	113.906689860	154.33884960
miles	0.020894530	0.03611557
hours	-6.281597236	1.97372814
passengers	0.829039697	1.02144670
delay	-0.009000196	0.06319060

11.40 R Interpretation

The regression model is statistically significant overall ($p < 0.001$), but it explains a modest amount of variation in coach prices ($R^2 = 0.1448$), meaning about 14.5% of the variation in price is captured by the model.

- **Miles** ($\beta = 0.0285$, $p < 0.001$)
Shows a significant positive effect, indicating that longer flights are associated with higher coach prices.
- **Passengers** ($\beta = 0.9252$, $p < 0.001$)
Has a strong positive effect, meaning flights with more passengers tend to have higher coach prices.
- **Hours** ($\beta = -2.15$, $p = 0.306$)
Not statistically significant after controlling for other variables.

- **Delay** ($\beta = 0.0271$, $p = 0.141$)
Also not a significant predictor of coach price.

11.41 PYTHON CODE

```
import pandas as pd
import statsmodels.api as sm
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
X = sample_df[['miles', 'hours', 'passengers', 'delay']]
X = sm.add_constant(X)      # Add intercept
y = sample_df['coach_price']
model = sm.OLS(y, X).fit()
print(model.summary())
```

11.42 PYTHON OUTPUT

OLS Regression Results

```
=====
Dep. Variable:          coach_price    R-squared:                0.145
Model:                  OLS           Adj. R-squared:            0.144
Method:                 Least Squares    F-statistic:              425.6
Date:                  Sun, 17 May 2026    Prob (F-statistic):       0.00
Time:                  23:12:56          Log-Likelihood:          -55838.
No. Observations:      10063            AIC:                    1.117e+05
Df Residuals:          10058            BIC:                    1.117e+05
Df Model:               4
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	145.7542	10.113	14.413	0.000	125.931	165.578
miles	0.0278	0.004	7.108	0.000	0.020	0.035
hours	-1.8957	2.119	-0.895	0.371	-6.049	2.258
passengers	0.8799	0.048	18.289	0.000	0.786	0.974
delay	0.0128	0.015	0.828	0.408	-0.018	0.043

```
=====
Omnibus:                291.071    Durbin-Watson:              1.994
Prob(Omnibus):           0.000    Jarque-Bera (JB):          302.594
Skew:                   -0.406    Prob(JB):                  1.96e-66
Kurtosis:                2.747    Cond. No.:                  3.64e+04
=====
```

11.43 Python Interpretation

The regression model is statistically significant overall ($F = 425.6$, $p < 0.001$), but has modest explanatory power with $R^2 = 0.145$. This means the model explains about 14.5% of the variation in coach prices.

Miles

Coefficient = 0.0278, $p < 0.001$

There is a significant positive relationship, indicating that longer distances increase coach prices.

Passengers

Coefficient = 0.8799, $p < 0.001$

Shows a strong and significant effect, meaning more passengers are associated with higher prices.

Hours

Coefficient = -1.8957, $p = 0.371$

Not statistically significant, indicating flight duration does not independently affect price.

Delay

Coefficient = 0.0128, $p = 0.408$

Not significant, showing that delays have no meaningful effect on pricing.

11.44 LOGISTIC REGRESSION

11.45 R CODE

```
df <- read.csv("flight_data.csv")
set.seed(10063)
sample_df <- df[sample(nrow(df), 10063), ]
sample_df$redeye_num <- ifelse(sample_df$redeye == "Yes", 1, 0)
sample_df$weekend_num <- ifelse(sample_df$weekend == "Yes", 1, 0)
model <- glm(redeye_num ~ miles + hours + passengers + delay + weekend_num,
             data = sample_df,
             family = binomial(link = "logit"))
summary(model)

cat("\n--- CORRECTED ODDS RATIOS & 95% CI (WALD) ---\n")
odds_ratios <- exp(coef(model))
conf_intervals <- exp(confint.default(model))
results_table <- cbind(Odds_Ratio = odds_ratios, conf_intervals)
print(results_table)
```

11.46 R OUTPUT

```
summary(model)
Call:
glm(formula = redeye_num ~ miles + hours + passengers + delay +
     weekend_num, family = binomial(link = "logit"), data = sample_df)

Coefficients:
```

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept) 169.298103  36.820288   4.598 4.27e-06 ***
miles       -0.004031   0.003532  -1.141  0.2538
hours        2.028547   1.919584   1.057  0.2906
passengers  -0.929865   0.202100  -4.601 4.20e-06 ***
delay        0.001376   0.009503   0.145  0.8849
weekend_num  1.803655   1.084327   1.663  0.0962 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 3881.769  on 10062  degrees of freedom
Residual deviance:  36.945  on 10057  degrees of freedom
AIC: 48.945

```

```

Number of Fisher Scoring iterations: 14
print(results_table)

```

```

              Odds_Ratio          2.5 %          97.5 %
(Intercept) 3.351442e+73 1.526673e+42 7.357283e+104
miles       9.959775e-01 9.891065e-01  1.002896e+00
hours       7.603032e+00 1.766177e-01  3.272950e+02
passengers  3.946070e-01 2.655442e-01  5.863985e-01
delay       1.001377e+00 9.828993e-01  1.020202e+00
weekend_num 6.071797e+00 7.249968e-01  5.085087e+01

```

11.47 R Interpretation

This logistic regression model predicts the probability that a flight is a redeye flight (redeyenum = 1).

The results show that the number of passengers is the only strong and statistically significant predictor. Its odds ratio (OR = 0.395, $p < 0.001$) indicates that as passengers increase, the odds of a flight being a red-eye decrease significantly by about 60% per unit increase in passengers.

The variable weekendnum suggests that flights on weekends may be more likely to be redeye flights (OR = 6.07), meaning the odds are higher compared to weekdays. However, this effect is only marginally significant ($p \approx 0.096$) and has a very wide confidence interval, so the result is uncertain.

The variables miles, hours, and delay are not statistically significant, indicating they do not have a reliable effect on whether a flight is a red-eye in this dataset.

Overall, the model suggests that passenger count is the dominant factor, while weekend timing may have a weak influence, and other predictors do not meaningfully contribute.

11.48 PYTHON CODE

```
import pandas as pd
```



```

import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
df = pd.read_csv('flight_data.csv')
np.random.seed(10063)
sample_df = df.sample(n=10063, random_state=10063).copy()
sample_df['redeye_num'] = sample_df['redeye'].map({'Yes': 1, 'No': 0})
sample_df['weekend_num'] = sample_df['weekend'].map({'Yes': 1, 'No': 0})
formula = 'redeye_num ~ miles + hours + passengers + delay + weekend_num'
model = smf.logit(formula, data=sample_df).fit()
print(model.summary())
print("\n" + "="*20 + " ODDS RATIOS & 95% CI " + "="*20)
params = model.params
conf_int = model.conf_int()
results_df = pd.concat([params, conf_int], axis=1)
results_df.columns = ['Odds Ratio', '2.5%', '97.5%']
results_df = np.exp(results_df)
print(results_df)

```

11.49 PYTHON OUTPUT

Optimization terminated successfully.

Current function value: 0.000753

Iterations 19

Logit Regression Results

=====						
Dep. Variable:	redeye_num	No. Observations:	10063			
Model:	Logit	Df Residuals:	10057			
Method:	MLE	Df Model:	5			
Date:	Mon, 18 May 2026	Pseudo R-squ.:	0.9962			
Time:	02:07:53	Log-Likelihood:	-7.5730			
converged:	True	LL-Null:	-2006.0			
Covariance Type:	nonrobust	LLR p-value:	0.000			
=====						
	coef	std err	z	P> z	[0.025	0.975]

Intercept	244.1917	84.365	2.894	0.004	78.838	409.545
miles	-0.0010	0.007	-0.140	0.889	-0.015	0.013
hours	0.5711	3.474	0.164	0.869	-6.238	7.380
passengers	-1.3396	0.460	-2.912	0.004	-2.241	-0.438
delay	-0.0526	0.113	-0.466	0.641	-0.274	0.169
weekend_num	0.1644	1.474	0.112	0.911	-2.725	3.054
=====						

Possibly complete quasi-separation: A fraction 0.99 of observations can be perfectly predicted. This might indicate that there is complete quasi-separation. In this case some parameters will not be identified.

===== ODDS RATIOS & 95% CI =====			
	Odds Ratio	2.5%	97.5%
Intercept	1.124914e+106	1.734145e+34	7.297151e+177
miles	9.990164e-01	9.852998e-01	1.012924e+00
hours	1.770280e+00	1.954395e-03	1.603511e+03
passengers	2.619454e-01	1.063216e-01	6.453569e-01
delay	9.488051e-01	7.606280e-01	1.183536e+00
weekend_num	1.178727e+00	6.555550e-02	2.119420e+01

11.50 Python Interpretation

11.51 Python INTERPRETATION

The model indicates that passengers is the only statistically significant predictor of whether a flight is a red-eye.

Passengers (coef = -1.34, OR ≈ 0.26 , $p = 0.004$):

A higher number of passengers significantly reduces the odds of a red-eye flight. Specifically, each unit increase in passengers is associated with about a 74% decrease in odds (since $1 - 0.26 \approx 0.74$). This is a strong and reliable effect.

Miles, Hours, Delay, Weekendnum:

All have p -values greater than 0.05, meaning there is no statistically reliable evidence that they affect red-eye probability in this model. Their confidence intervals are also wide, indicating high uncertainty.